UNIVERSITY OF BARISAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

3rd Year 2nd Semester Final Examination

Course Title: Peripherals and Interfacing
Course Code: CSE-3207 Admission Session: 2021-22

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings.

1. a) What is Interface? Briefly explain different types of Interfaces. Differentiate between off-line and on-line peripherals [4]

   b) What do you mean by Computer peripheral? List down the input and output Peripherals. [4]

   c) A university plans to build a modern computer laboratory. Recommend suitable peripheral devices for student programming, online classes, multimedia presentations, and document digitization. Justify your choices. [4]

2. a) Describe the working principle of a keyboard. Explain how a key press is detected using the keyboard matrix concept. [4]

   b) Compare membrane, mechanical, and wireless keyboards. Discuss advantages, disadvantages, and applications. [4]

   c) A software company has programmers, graphic designers, and data-entry operators. Recommend suitable keyboards for each and justify your answer. [4]

3. a) Explain the working principle of an optical mouse. How does it detect movement compared with a mechanical mouse? [4]

   b) Differentiate among laser, optical, and wireless mice in terms of accuracy, power consumption, cost, and applications. [4]

   c) A hospital wants computers in operating rooms where hygiene is important. Recommend suitable pointing devices and justify your answer. [4]

4. a) Explain the working principles of Inkjet Printer and Laser Printer with neat diagrams. [4]

   b) Compare Inkjet, Laser, and Dot Matrix printers using printing speed, print quality, cost per page, maintenance, and applications. [4]

   c) Recommend a printer for a publishing company that prints thousands of pages daily. Justify your answer. [4]

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5. a) Explain how an LCD/LED monitor produces images. Describe the importance of resolution, refresh rate, response time, and brightness. [4]

b) Differentiate between LCD, LED, and OLED monitors. [4]

c) Recommend suitable monitors for a professional gamer and a graphic designer with [4]

6. a) Explain the working principles of a scanner and a barcode reader. How do they convert physical information into digital data? [4]

b) Differentiate between flatbed scanner, handheld scanner, and barcode reader based on speed, accuracy, cost, and applications. [4]

c) Recommend suitable devices for a supermarket and a university library for automatic identification. [4]

7. a) Explain the functions and working principles of microphone, speaker, and headphone. [4]

c) Differentiate between dynamic and condenser microphones. Also compare wired and Bluetooth speakers. [4]

d) Recommend suitable audio devices for an online lecturer considering sound quality, noise reduction, portability, and cost. [4]

8. a) Explain the purposes and working principles of webcam, projector, and USB. [4]

b) Compare Webcam vs Digital Camera and USB 2.0 vs USB 3.0 vs USB Type-C in terms of speed, power delivery, and applications. [4]

c) A university plans to establish a smart classroom for hybrid teaching. Select appropriate combinations of webcam, projector, and USB devices and explain how they work together. [4]

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